

Building an Explainable Institutional Reasoning System

A Prototype for Strategic Academic Decision Support

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Our original discussion began with a simple question:

Question

Can artificial intelligence become an active participant in strategic academic planning while remaining transparent, explainable, and defensible?

This project explores one possible answer.

The Challenge

Universities routinely make strategic decisions involving

- enrollment trends
- academic program development
- faculty hiring
- accreditation
- budgeting
- long-term strategic planning

How these decisions are often made

They rely heavily upon collective human knowledge accumulated over many years—what we call **institutional memory**.

That memory is enormously valuable, but it is difficult to preserve, search, update, and transfer.

Traditional Decision Support

Institutional Memory

- Experienced administrators
- Experienced faculty and staff
- Committee deliberation
- Historical knowledge
- Informal relationships

Supporting Materials

- Reports
- Spreadsheets
- Policies
- Accreditation records
- Planning documents

The institution possesses substantial knowledge, but much of it is distributed across people, files, systems, and historical records.

Every new question can therefore require reconstructing the relevant institutional context.

Rather than producing another report...

Build an Institutional Reasoning System

that continuously

- organizes institutional knowledge,
- retrieves relevant evidence,
- evaluates scenarios,
- produces explainable recommendations,
- and supports—rather than replaces—human judgment.

Design Philosophy

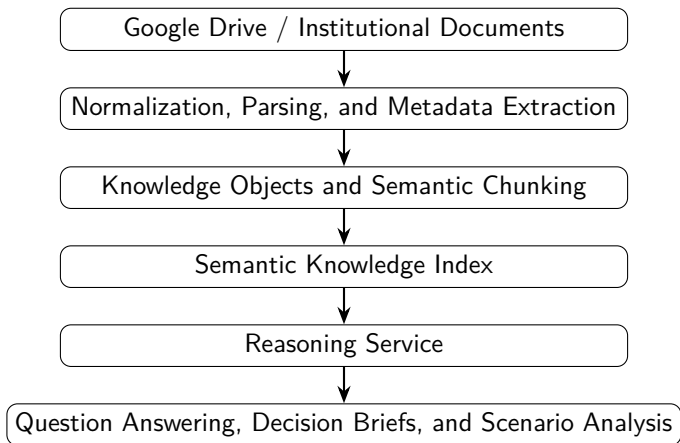
The objective is **not** automated decision making.

The objective is to augment human decision making with transparent, evidence-based reasoning.

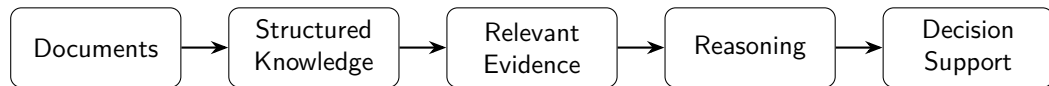
Every recommendation should be

- Explainable
- Defensible
- Traceable to institutional evidence
- Reproducible
- Open to human review

Prototype Architecture



From Documents to Institutional Knowledge



The system does not ask a language model to read every institutional document for every question.

It first identifies the small body of evidence most relevant to the specific decision, and then asks the reasoning service to analyze it.

An Important Architectural Principle

Knowledge objects store facts.

Documents, metadata, source locations, semantic chunks, timestamps, and provenance.

Services derive meaning.

Retrieval, ranking, synthesis, scenario evaluation, decision support, and institutional reasoning.

Knowledge objects store facts. Services derive meaning.

This separation allows the same institutional knowledge layer to support multiple applications without embedding conclusions or recommendations permanently into the underlying data.

Why Explainability Matters

Traditional AI systems often produce answers without showing how they were obtained.

For institutional decision support, that is unacceptable.

Every recommendation should answer three questions:

- ① What evidence was used?
- ② Why was that evidence selected?
- ③ Could another person independently reproduce the result?

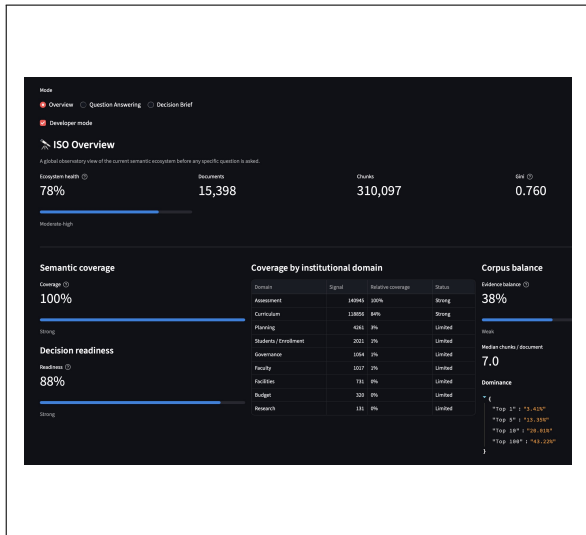
Our architecture was designed around these questions from the very beginning.

The Semantic Observatory

The prototype continuously monitors the health of the semantic ecosystem.

Examples include

- document counts
- parser statistics
- chunk distributions
- corpus dominance
- outlier detection
- parser quality
- retrieval benchmarking



Why Monitor Semantic Health?

The quality of an AI system depends fundamentally on the quality of the information ecosystem it reasons over.

Example

One corrupted spreadsheet generated nearly

50,000

semantic chunks.

Cleaning the spreadsheet reduced this to only a few hundred chunks and improved the overall balance of the corpus.

The lesson is simple:

Healthy reasoning requires a healthy semantic ecosystem.

Semantic Health Metrics

Current prototype corpus

Documents	~15,400
Semantic Chunks	~310,000
Median Chunks / Document	7
90th Percentile	24
99th Percentile	147
Corpus Gini Coefficient	0.76

The Gini coefficient provides a quantitative measure of whether knowledge is reasonably distributed across the corpus or dominated by only a handful of documents.

Developer Diagnostics

Every question can be inspected in detail.

For every query we record

- FAISS retrieval
- deduplication
- reranking
- thresholding
- final evidence supplied to the LLM
- execution timing

This transforms the retrieval pipeline from a "black box" into an engineering system that can be measured, tested, and improved.

The Working Prototype

The current system already supports three distinct modes.

Question Answering

Retrieve evidence and answer factual questions with citations.

Semantic Observatory

Monitor the health, balance, and structure of the semantic ecosystem.

Decision Briefs

Synthesize evidence across multiple domains for strategic questions.

All three capabilities use the same underlying institutional knowledge layer.

The demonstration will show three increasingly sophisticated uses.

① **Question answering**

What is the CNU travel reimbursement policy?

② **Semantic Health**

How do we know the information environment is reliable?

③ **Decision support**

What additional resources would be required to support a Mechanical Engineering major?

From Question Answering to Decision Support

A factual question usually has a relatively direct answer.

Question Answering

What is the current travel reimbursement policy?

A strategic question requires synthesis across many sources.

Decision Support

What additional resources would be required to launch a Mechanical Engineering major?

No single document contains the answer.

The answer emerges from the institutional knowledge ecosystem.

Mechanical Engineering Decision Brief

The system decomposes the question into several evidence domains.

- Existing faculty expertise
- Required faculty expertise
- Number of faculty lines
- Existing curriculum
- New course requirements
- Laboratory space
- Equipment needs
- ABET requirements
- Enrollment assumptions
- Budget implications

The resulting brief identifies evidence, likely gaps, risks, uncertainties, and missing information.

What a Decision Brief Should Contain

- 1 Decision question
- 2 Executive summary
- 3 Relevant institutional evidence
- 4 Existing strengths
- 5 Resource gaps
- 6 Competing considerations
- 7 Risks and uncertainties
- 8 Missing information
- 9 Supporting sources

The system does not make the decision.

It structures the evidence needed for a defensible human decision.

Why This Is More Than a Chatbot

The language model is not the system.

It is one reasoning service operating over carefully engineered institutional knowledge.

The system also includes

- corpus stewardship,
- provenance,
- retrieval,
- benchmarking,
- observability,
- validation,
- and human governance.

What We Have Learned

The hardest problem was not the language model.

The most difficult and important work involved

- determining what belongs in the corpus,
- preserving provenance,
- detecting corrupted, duplicated, or dominant documents,
- measuring retrieval quality,
- defining meaningful benchmarks,
- and making the system observable.

Trustworthy AI depends upon trustworthy information infrastructure.

Prototype Status

The current prototype demonstrates that the architecture is practical and technically feasible.

Knowledge Infrastructure

- Document normalization
- Semantic chunk generation
- Semantic indexing
- Semantic Observatory

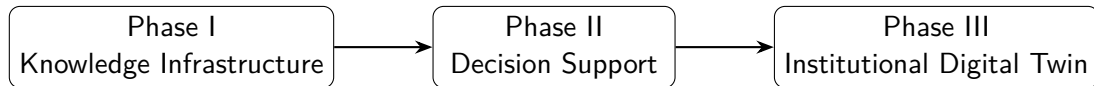
Reasoning and Evaluation

- Retrieval benchmarking
- Explainable diagnostics
- Question answering
- Decision Brief generation

Current transition

The focus is shifting from proving technical feasibility toward developing a sustained institutional decision-support capability.

The Road Ahead



Today's prototype completes much of the first phase and begins to demonstrate the second.

Near-Term Opportunities

Examples that could be explored during the coming months include

- Mechanical Engineering program planning
- Faculty hiring scenarios
- Enrollment forecasting
- Space utilization
- Laboratory planning
- Accreditation support
- Strategic budgeting
- Institutional policy analysis

The same semantic infrastructure supports each of these applications.

An Emerging View of AI

One lesson has become increasingly clear during development.

Traditional View

AI is a model that answers questions.

Our Working View

AI is one reasoning component operating within a carefully engineered information ecosystem.

The quality of reasoning is often limited more by the quality of the semantic ecosystem than by the reasoning model itself.

Digital Twin

A long-term objective is to develop a semantic representation of the institution itself.

Rather than storing only documents,
the system gradually develops an interconnected representation of

- people
- programs
- curricula
- facilities
- policies
- finances
- strategic initiatives
- institutional relationships

This evolving semantic model provides the foundation for institutional reasoning.

One Observation

During development we discovered something unexpected.

The engineering required to create trustworthy institutional knowledge turned out to be substantially more difficult than connecting a language model.

This project has therefore become as much about **knowledge engineering** as it is about artificial intelligence.

- Explainable AI for higher education is achievable.
- Institutional knowledge can be organized into a reusable semantic infrastructure.
- Trust depends upon explainability, provenance, and continuous measurement.
- A healthy semantic ecosystem is essential for trustworthy institutional reasoning.

The objective is not to automate
university decision making.

The objective is to help people make better-informed, more transparent,
and more defensible decisions.

Thank you.